

FINAL EXAM

ECON301—ECONOMETRICS I

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1. The Racoon Memory Chip Co. (RMC for short) instructs its resident econometrician to investigate the operating-life of their new 36 gigabyte memory module in order to provide product information for potential buyers. The population distribution of operating lives can be modeled as a random variable having an exponential distribution, as follows:

$$X \sim f(x; \theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}} I_{(0, \infty)}(x),$$

where θ is a *parameter* in the above distribution. The econometrician intends to draw a random sample of $n = 10,000$ of memory modules from RMC's production and apply a non-destructive test that will determine each module's operating life. She then intends to use the outcome of the random sample to provide estimates of both the mean and the variance of the module's operating life. She needs your help in answering a few questions.

The outcome of the random sample resulted in the following summary statistics:

$$\frac{1}{n} \sum_{i=1}^n x_i = 10.03702, \quad \frac{1}{n} \sum_{i=1}^n x_i^2 = 199.09634.$$

Note: the operating life is measured in 1,000 h units.

- (a) (10%) Show that $E(X) = \theta$ and that $\text{Var}(X) = \theta^2$.

If you could not show the results in (a), take them as facts for the remainder of the problem.

- (b) (5%) Consider the sample mean $\bar{X}_n$. Is $\bar{X}_n$ a consistent estimator of θ ?

- (c) (5%) Show that $\sqrt{n}(\bar{X}_n - \theta) \rightarrow_d N(0, \theta^2)$.

In order to estimate the variance in the asymptotic distribution above, the econometrician is considering two candidate estimators: $\bar{X}_n^2$ (the square of the sample mean), and $S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$.

- (d) (15%) Does $E(\bar{X}_n^2) = \theta^2$? Does $E(S_n^2) = \theta^2$? Are these two estimators unbiased? *Hint: expand the square in S_n^2 and simplify. When taking the expected values, remember to use the variance calculation formula.*

- (e) (10%) Is $\bar{X}_n^2$ consistent? Is S_n^2 consistent?

- (f) (10%) Using what you learned in (d) and (e), why is it okay to use either $\bar{X}_n^2$ or S_n^2 to estimate the variance? Now calculate the outcomes of both $\bar{X}_n^2$ and S_n^2 .

- (g) (15%) The econometrician does not like that both $\bar{X}_n^2$ and S_n^2 are biased. She wants to define new estimators $T_n = a_n \bar{X}_n^2$ and $W_n = b_n S_n^2$ for appropriate sequences of numbers (a_n) and (b_n) so that $E(T_n) = E(W_n) = \theta^2$ for all n . Can she do it? How? If she (and you) can, then use the appropriate outcomes of T_n and W_n to provide new estimates for θ^2 . Is it true that T_n and W_n are both consistent?

- (h) (15%) Using any of the estimators for the variance discussed above, provide the details to build an asymptotic pivot to make inference on θ .

- (i) (10%) Construct an asymptotic 1% confidence interval for θ .

- (j) (5%) Using the sample mean to give an estimate for θ , compute the probability that the operating-life of a memory module will be greater than 20,000 hours.

2. In this problem, you have a data set consisting of ZIP code-level data on prices for various items at fast-food restaurants, along with characteristics of the zip code population, in New Jersey and Pennsylvania.¹ Suppose you are considering to estimate the following best linear predictor:

$$E^*(lpsoda|prpbck, lincome, prppov) = \beta_0 + \beta_1 prpbck + \beta_2 lincome + \beta_3 prppov, \quad (1)$$

where *lpsoda* is the log of the price of a medium soda, *prpbck* is the proportion of the population that is black, *lincome* is the log of median family income, and *prppov* is the proportion of the population in poverty. The goal is to see whether fast-food restaurants charge higher prices in areas with a larger concentration of blacks.

- (a) (5%) What is the interpretation of the coefficient β_1 ?
- (b) (5%) What is the interpretation of the coefficient β_2 ?
- (c) (5%) Use the results presented in column (1) of table 1 to test if $\hat{\beta}_1$ is statistically different from zero at the 5% level against a two-sided alternative. What about at the 1% level?
- (d) (10%) Is *lincome* statistically significant? Is *prppov* statistically significant? Test each of these two individual hypothesis at the 5% and 1% levels.
- (e) (15%) Test for the global significance of model (1). You will need to write the restriction matrix, write the null and the alternative hypothesis in terms of your restrictions matrix, propose the test statistic and its distribution, and then, simplify the expression for the test statistic as much as you can without actually computing it. The actual computed value of the appropriate test statistic appears in table 1. Use it to test the global significance of the model at the 5% and 1% levels.
- (f) (5%) Now suppose that you are considering adding the variable *lhseval*, which is the log of the median housing value, to the regression, i.e., you plan to estimate the following BLP:

$$E^*(lpsoda|prpbck, lincome, prppov, lhseval) = \beta_0 + \beta_1 prpbck + \beta_2 lincome + \beta_3 prppov + \beta_4 lhseval. \quad (2)$$

The estimation results for this model appear in column (2) in table 1. Interpret the coefficient β_4 and test its individual significance at the 5% and 1% levels.

- (g) (15%) Test for the global significance of model (2). Similarly to task (e), write all the details to perform this test. Simplify the test statistic as much as you can without actually computing it. The actual computed value of the appropriate test statistic appears in table 1. Use it to test the global significance of the model at the 5% and 1% levels.
- (h) (10%) In the regression for model (2), what happens to the individual statistical significance of *lincome* and *prppov*?
- (i) (15%) Are *lincome* and *prppov* jointly significant? Write out all the details to carry out this test: restriction matrix, null and alternatives, test statistic and its distribution, etc. Simplify the test statistic as much as you can, without actually computing it. The observed value for the test statistic for this test is: 6.564999. Use it to test at the 5% and 1% levels.
- (j) (15%) Given the results of models (1) and (2), which one would you report as being the most reliable in determining whether the racial makeup of a zip code influences local fast-food prices? What do you conclude from this exercise?

3. Using a sample of recent high school graduates in the US, you are interested in estimating the following BLP:

$$E^*[sat|hsiz, hsize^2, female, black, female \times black] = \beta_0 + \beta_1 hsize + \beta_2 hsize^2 + \beta_3 female + \beta_4 black + \beta_5 female \times black \quad (3)$$

where the variables are: *sat* is the combined SAT score; *hsiz* is the size of the student's high school graduating class, in hundreds; *female* is a gender dummy variable; *black* is a race dummy variable equal to one for blacks, and zero otherwise.

The estimation results are in Table 2. The standard errors are in parenthesis, below the estimated coefficient. The estimate for the variance covariance matrix is provided in table 3.

- (a) (15%) Is there strong evidence that $hsize^2$ should be included in the model? From this equation, what is the optimal high school size?
- (b) (15%) Holding *hsiz* fixed, what is the estimated difference in SAT score between nonblack females and nonblack males? Construct a hypothesis test in which the null hypothesis is that there is no difference between nonblack females and nonblack males vs the alternative that there is a difference. Test it at the 1% level.

¹Source: Wooldridge data set DISCRIM, taken from K. Graddy (1997), "Do Fast-Food Chains Price Discriminate on the Race and Income Characteristics of an Area?" Journal of Business and Economic Statistics 15, 391-401.

- (c) (15%) What is the estimated difference in SAT score between nonblack males and black males? Test the null hypothesis that there is no difference between their scores, against the alternative that there is a difference at the 5% level.
- (d) (25%) What is the estimated difference in SAT score between black females and nonblack females? Construct the relevant hypothesis test to test whether this difference is significant or not and test it at the 1% level.
- (e) (30%) Consider student A , which graduated from a high school class of 400 ($hsize = 4$), is *female and black*. Student B graduated from a high school class of 500 ($hsize = 5$) and is also *female and not black*. Let $\Delta sat = sat_B - sat_A$ denote the difference in SAT scores where sat_B and sat_A are the SAT scores of students B and A respectively. Delineate a procedure that will allow you to construct a 95% confidence interval around Δsat . Be precise and detailed in describing your procedure. For an appropriately defined vector λ for this problem, you may find useful the following computation $\lambda^\top \hat{\Sigma} \lambda = 136.5755$.

Bonus Question: What is Prof. Alejandro's favorite animal? (3 points if you name the animal + 2 points if you name it in Spanish.)

Table 1: Estimation results for problem 2

	Dependent variable:	
	lpsoda	
	(1)	(2)
prpblck	0.073** (0.035)	0.098*** (0.030)
lincome	0.137*** (0.026)	-0.053 (0.035)
prppov	0.380*** (0.135)	0.052 (0.125)
lhseval		0.121*** (0.017)
Constant	-1.463*** (0.285)	-0.842*** (0.271)
Observations	401	401
R ²	0.087	0.184
Adjusted R ²	0.080	0.176
Global significance Wald statistic	39.35043	104.145
Note:	*p<0.1; **p<0.05; ***p<0.01	

Table 2: Estimation results from the model (3)

<i>Dependent variable:</i>	
	sat
hsize	19.297*** (3.814)
hsizesq	−2.195*** (0.524)
female	−45.091*** (4.226)
black	−169.813*** (15.423)
female×black	62.306*** (19.305)
Constant	1,028.097*** (6.355)
Observations	4,137
R ²	0.086
Adjusted R ²	0.085
Note:	*p<0.1; **p<0.05; ***p<0.01

Table 3: Estimated Variance-Covariance Matrix

$$\hat{\Omega} = \begin{bmatrix} 40.386516 & -20.199194 & 2.335095 & -9.031893 & -11.536296 & 8.210525 \\ -20.199194 & 14.545411 & -1.900289 & 0.078404 & 1.261833 & 0.068763 \\ 2.335095 & -1.900289 & 0.274387 & -0.000038 & -0.077895 & 0.031187 \\ -9.031893 & 0.078404 & -0.000038 & 17.863001 & 8.820194 & -17.816754 \\ -11.536296 & 1.261833 & -0.077895 & 8.820194 & 237.863505 & -237.367652 \\ 8.210525 & 0.068763 & 0.031187 & -17.816754 & -237.367652 & 372.702161 \end{bmatrix}$$

NORMAL CUMULATIVE DISTRIBUTION FUNCTION

x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

CHI-SQUARED PERCENTAGE POINTS (Cumulative Distribution Function)

ν	60.0%	66.7%	75.0%	80.0%	87.5%	90.0%	95.0%	97.5%	99.0%	99.5%	99.9%
1	0.708	0.936	1.323	1.642	2.354	2.706	3.841	5.024	6.635	7.879	10.828
2	1.833	2.197	2.773	3.219	4.159	4.605	5.991	7.378	9.210	10.597	13.816
3	2.946	3.405	4.108	4.642	5.739	6.251	7.815	9.348	11.345	12.838	16.266
4	4.045	4.579	5.385	5.989	7.214	7.779	9.488	11.143	13.277	14.860	18.467
5	5.132	5.730	6.626	7.289	8.625	9.236	11.070	12.833	15.086	16.750	20.515
6	6.211	6.867	7.841	8.558	9.992	10.645	12.592	14.449	16.812	18.548	22.458
7	7.283	7.992	9.037	9.803	11.326	12.017	14.067	16.013	18.475	20.278	24.322
8	8.351	9.107	10.219	11.030	12.636	13.362	15.507	17.535	20.090	21.955	26.125
9	9.414	10.215	11.389	12.242	13.926	14.684	16.919	19.023	21.666	23.589	27.877
10	10.473	11.317	12.549	13.442	15.198	15.987	18.307	20.483	23.209	25.188	29.588
11	11.530	12.414	13.701	14.631	16.457	17.275	19.675	21.920	24.725	26.757	31.264
12	12.584	13.506	14.845	15.812	17.703	18.549	21.026	23.337	26.217	28.300	32.910
13	13.636	14.595	15.984	16.985	18.939	19.812	22.362	24.736	27.688	29.819	34.528
14	14.685	15.680	17.117	18.151	20.166	21.064	23.685	26.119	29.141	31.319	36.123
15	15.733	16.761	18.245	19.311	21.384	22.307	24.996	27.488	30.578	32.801	37.697
16	16.780	17.840	19.369	20.465	22.595	23.542	26.296	28.845	32.000	34.267	39.252
17	17.824	18.917	20.489	21.615	23.799	24.769	27.587	30.191	33.409	35.718	40.790
18	18.868	19.991	21.605	22.760	24.997	25.989	28.869	31.526	34.805	37.156	42.312
19	19.910	21.063	22.718	23.900	26.189	27.204	30.144	32.852	36.191	38.582	43.820
20	20.951	22.133	23.828	25.038	27.376	28.412	31.410	34.170	37.566	39.997	45.315
21	21.991	23.201	24.935	26.171	28.559	29.615	32.671	35.479	38.932	41.401	46.797
22	23.031	24.268	26.039	27.301	29.737	30.813	33.924	36.781	40.289	42.796	48.268
23	24.069	25.333	27.141	28.429	30.911	32.007	35.172	38.076	41.638	44.181	49.728
24	25.106	26.397	28.241	29.553	32.081	33.196	36.415	39.364	42.980	45.559	51.179
25	26.143	27.459	29.339	30.675	33.247	34.382	37.652	40.646	44.314	46.928	52.620
26	27.179	28.520	30.435	31.795	34.410	35.563	38.885	41.923	45.642	48.290	54.052
27	28.214	29.580	31.528	32.912	35.570	36.741	40.113	43.195	46.963	49.645	55.476
28	29.249	30.639	32.620	34.027	36.727	37.916	41.337	44.461	48.278	50.993	56.892
29	30.283	31.697	33.711	35.139	37.881	39.087	42.557	45.722	49.588	52.336	58.301
30	31.316	32.754	34.800	36.250	39.033	40.256	43.773	46.979	50.892	53.672	59.703
35	36.475	38.024	40.223	41.778	44.753	46.059	49.802	53.203	57.342	60.275	66.619
40	41.622	43.275	45.616	47.269	50.424	51.805	55.758	59.342	63.691	66.766	73.402
45	46.761	48.510	50.985	52.729	56.052	57.505	61.656	65.410	69.957	73.166	80.077
50	51.892	53.733	56.334	58.164	61.647	63.167	67.505	71.420	76.154	79.490	86.661
55	57.016	58.945	61.665	63.577	67.211	68.796	73.311	77.380	82.292	85.749	93.168
60	62.135	64.147	66.981	68.972	72.751	74.397	79.082	83.298	88.379	91.952	99.607